The Role of Artificial Intelligence in Autonomous Robots

Mini Project Report – Robotics & Automation

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Abstract — This report examines the central role that artificial intelligence (AI) plays in autonomous robots. It distinguishes autonomy from conventional automation, sets out the ‘sense–think–act’ architecture on which autonomous systems are built, and surveys the main AI techniques involved: perception, localisation and mapping, planning and decision-making, learning, and control. It then reviews how these capabilities are being used today — in self-driving vehicles, warehouse fleets, and the new generation of humanoid robots — before discussing the key challenges of safety, data, generalisation, efficiency and trust, and the likely future direction of the field around the convergence of foundation models and robotics.

1. Introduction

For most of the twentieth century, robots were powerful but rigid: industrial arms that repeated a fixed motion thousands of times a day inside a safety cage, and conveyor lines controlled by programmable logic controllers in tightly structured factories. Such machines are highly automated, but they are not autonomous — they cannot cope with anything their programmer did not anticipate. Artificial intelligence (AI) is what closes that gap. By giving a machine the ability to perceive its surroundings, build an understanding of them, decide what to do and learn from experience, AI turns a pre-programmed device into an autonomous robot that can operate in messy, unpredictable, real-world environments.

This report examines the role AI plays in autonomous robots. It explains what separates autonomy from automation, surveys the main families of AI techniques used, reviews current real-world applications, and discusses the key challenges and likely future directions. The period around 2025–26 is widely seen as a turning point: the same foundation-model methods that transformed language and vision are now being applied directly to physical action, a shift the industry has begun to call ‘physical AI’.

2. From Automation to Autonomy

An autonomous robot is one that can achieve a goal in an uncertain environment with little or no human intervention. Autonomy is best understood as a spectrum rather than a single property. At one end sits teleoperation, where a human controls every movement; then assisted and conditional autonomy, where the robot handles routine operation but a person supervises and can intervene; and at the far end full autonomy, where the robot manages the task end to end. The automotive industry formalises this idea in the SAE levels of driving automation (Levels 0 to 5).

Most autonomous systems are organised around a continuous ‘sense–think–act’ loop, shown in Figure 1. Sensors gather raw data; perception interprets it; localisation and mapping place the robot within the world; a planner decides what to do; controllers convert the plan into commands; and actuators move the robot, changing the world and producing new observations. AI contributes at every stage, and a learning component continually improves the others. The contrast with classical automation — such as a PLC-driven conveyor that follows fixed logic in a structured setting — is that an autonomous robot must reason about situations that were never explicitly programmed.

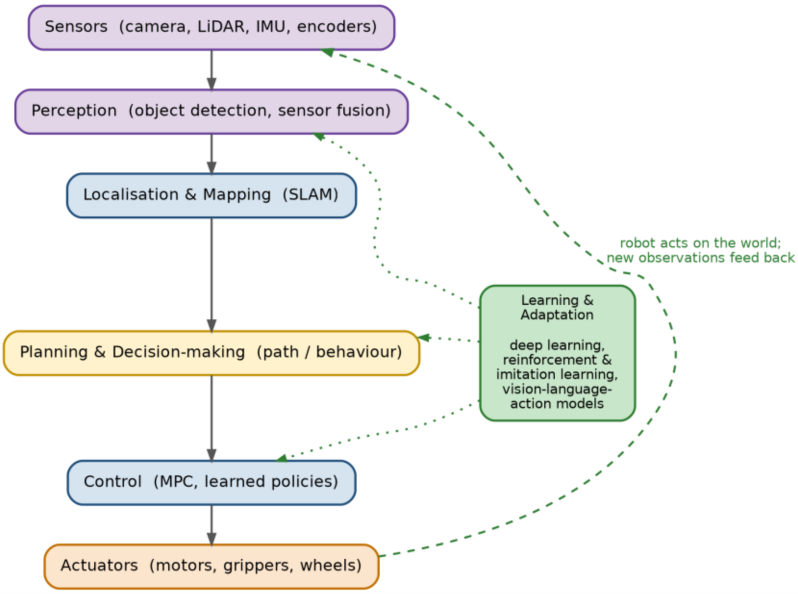

Figure 1 – The 'sense–think–act' pipeline of an autonomous robot. AI contributes at every stage, with a learning component that continually improves perception, planning and control.

3. Core AI Techniques in Autonomous Robots

3.1 Perception and sensor fusion

Perception is the robot's ability to turn raw sensor data into a usable model of the world. Modern perception relies heavily on deep learning: convolutional neural networks and, increasingly, vision transformers detect, classify and segment objects in camera images, while related models process LiDAR point clouds and radar returns. Because no single sensor is reliable in all conditions, autonomous robots use sensor fusion to combine cameras (rich detail), LiDAR (accurate depth) and radar (robust in poor weather) into a single, more trustworthy estimate of their surroundings.

3.2 Localisation and mapping

To act sensibly, a robot must know where it is. Simultaneous Localisation and Mapping (SLAM) lets a robot build a map of an unknown environment while at the same time tracking its own position within that map, typically by fusing visual and inertial data (visual-inertial odometry). Self-driving cars often combine SLAM with pre-built high-definition (HD) maps, while warehouse and home robots rely on real-time SLAM to navigate spaces that change from day to day.

3.3 Decision-making and motion planning

Once a robot understands its situation, it must decide how to act. Motion planning computes a collision-free path to a goal using classical algorithms such as A* and Rapidly-exploring Random Trees (RRT), while higher-level behaviour planning chooses between options — overtake or follow, pick or place. Because the world is uncertain and only partially observable, these decisions are increasingly framed as planning under uncertainty, for example using partially observable Markov decision processes (POMDPs), or solved with learned policies that map situations directly to actions.

3.4 Learning

Learning is what allows autonomous robots to improve and to handle situations their designers never foresaw. Supervised deep learning trains the perception system from large labelled datasets. Reinforcement learning (RL) lets a robot discover control strategies — such as a legged robot's gait — through trial and error against a reward signal, while imitation learning teaches skills from human demonstrations. Much of this training happens in simulation, after which the challenge of sim-to-real transfer is to make policies learned in a simulator work on real hardware.

The most significant recent development is the arrival of robot foundation models, and in particular vision-language-action (VLA) models. Building on large language and vision models, a VLA takes in what the robot sees together with a natural-language instruction and outputs low-level actions, allowing a single model to perform many tasks and to generalise to new ones. Notable examples include Google's RT-2 and Gemini Robotics, the open OpenVLA and SmolVLA models, Physical Intelligence's π0 family, NVIDIA's GR00T N1 for humanoids, and Figure's Helix. Shared datasets such as Open X-Embodiment, which pool demonstrations from many different robots, have helped these generalist models emerge.

3.5 Control

Control turns decisions into smooth, stable motion. Classical controllers such as PID loops and model-predictive control (MPC) remain widely used and are valued for their predictability, but they are increasingly combined with learned controllers. Whole-body control of legged and humanoid robots — where dozens of joints must be coordinated to keep balance — is one area where reinforcement learning has produced striking results.

3.6 Human–robot interaction

Finally, AI is making robots easier to work with. Large language models now allow people to instruct robots in plain language rather than code; Amazon's latest warehouse robot, for example, can be told what to do in conversational text and works out the route and timing itself. Natural, safe human–robot interaction is essential as robots move out of cages and onto shared factory floors, pavements and homes. Figure 2 summarises the four core AI capability areas described above.

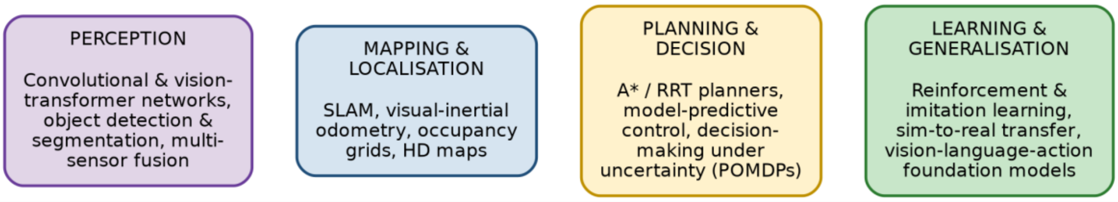

Figure 2 – The four core AI capability areas that underpin autonomous robots, with representative techniques.

4. Applications and Case Studies

AI-driven autonomy has moved rapidly from the laboratory into commercial use. Table 1 summarises the main application domains, together with representative systems and the AI techniques behind them.

Domain	Representative systems (2025–26)	Key AI techniques
Autonomous vehicles / robotaxis	Waymo (driverless in ~11 US cities; London & Tokyo next); Tesla Robotaxi; Wayve	Camera–LiDAR–radar fusion, deep perception, HD maps, motion planning, end-to-end learning
Warehouse & logistics	Amazon (1M+ robots: Proteus, Sparrow, Cardinal; DeepFleet fleet AI); Agility Digit	Computer vision, grasp planning, reinforcement learning, multi-robot coordination

Humanoid robots	Figure 02/03 (BMW); Tesla Optimus; Boston Dynamics Atlas; Unitree; 1X NEO	Vision-language-action models, whole-body RL control, imitation learning
Aerial robots (drones)	Inspection, delivery, mapping & agricultural UAVs	Visual SLAM, obstacle avoidance, autonomous navigation, object detection
Healthcare & surgery	Robot-assisted surgical systems (e.g. da Vinci); service robots	Computer vision, sensor fusion, motion planning, haptic feedback
Agriculture & field	Autonomous tractors and crop-monitoring robots; planetary rovers (e.g. Perseverance)	Vision-based perception, GPS/SLAM navigation, fault-tolerant planning

Table 1 – Application domains for autonomous robots, with representative systems and key AI techniques.

Autonomous vehicles are the most visible example. By 2026 Waymo was running fully driverless ride-hailing in around a dozen US cities and preparing to launch in London and Tokyo; the company reports that across well over a hundred million driverless miles it has recorded roughly a 90% reduction in serious-injury crashes compared with human drivers, although regulators have also opened investigations into specific incidents. Other firms, including Tesla and the UK's Wayve, pursue a camera-only, end-to-end learning approach.

In warehousing and logistics, Amazon now operates more than a million robots — among them the autonomous mobile robot Proteus and vision-guided picking arms such as Sparrow — coordinated by a generative-AI system called DeepFleet. Such deployments show AI handling perception, grasping and large-scale multi-robot coordination together.

Humanoid robots are the fastest-moving frontier. Figure's robots spent about eleven months on BMW's production line in Spartanburg, contributing to more than 30,000 vehicles, while Tesla, Boston Dynamics, Agility, Unitree and others race to deploy general-purpose humanoids — most of them now driven by VLA-style models. Beyond these, autonomous drones inspect infrastructure and survey crops, robot-assisted surgical systems extend a surgeon's precision, autonomous tractors work fields, and planetary rovers navigate terrain millions of kilometres from Earth.

5. Challenges and Limitations

Despite rapid progress, important challenges remain.

- Safety and reliability: an autonomous robot must behave correctly across a vast 'long tail' of rare situations, and proving that it will is far harder than demonstrating it once. Rigorous verification, validation and certification of learning-based systems is still an open problem.
- Data and the sim-to-real gap: unlike text or images, physical-world data is expensive to collect, and policies trained in simulation often behave differently on real hardware.
- Generalisation and robustness: models can fail when conditions drift away from their training data, for example in unusual weather or lighting.
- Compute and energy: running large perception and planning models in real time on a moving, battery-powered robot is demanding — today's humanoids, for instance, typically operate for only a few hours between charges.
- Trust, security and ethics: learned systems can be opaque and vulnerable to adversarial attacks, while wider deployment raises questions of privacy, accountability and the effect on jobs. Amazon argues that its robots augment workers and create new technical roles, but unions and policymakers continue to debate displacement, and regulation for autonomous machines is still catching up.

6. Future Directions

Several trends are likely to shape the next few years. The clearest is the convergence of foundation models with robotics: general-purpose VLA models and 'world models' promise robots that can be instructed in language and adapt to new tasks without bespoke programming. Hardware is becoming cheaper and more capable, with humanoid robots beginning to reach genuine production volumes. Better simulators and large pools of synthetic and shared data — together with fleet learning, in which many robots learn from one another's experience — are accelerating training, while more efficient on-device models are bringing capable AI onto the robot itself. Alongside these technical advances, the development of safety standards, assurance methods and natural human–robot collaboration will be decisive in turning impressive demonstrations into dependable, everyday systems.

7. Conclusion

Artificial intelligence is the defining ingredient of the modern autonomous robot. It supplies the perception, mapping, planning, control and learning that allow a machine to act sensibly in an unstructured world, transforming the rigid automation of the past into adaptable, general-purpose systems. The progress seen in autonomous vehicles, warehouse fleets and humanoid robots shows how quickly these capabilities are maturing, while the remaining challenges — safety, data, generalisation, efficiency and trust — make clear that the field is still early in its journey. As foundation-model AI and capable, affordable hardware continue to converge, autonomous robots are set to become an increasingly visible part of industry and everyday life.

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